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Employee Burnout Prediction

Project Report

SUBMITTED TO:

Dr. Madhu Bhan

(Associate Professor)

(AIIT)

SUBMITTED BY:

Krishnam Mavani

(MSc Data Science)

(A869117725001)

Amity University Bengaluru

NH-207, opposite to the office of Deputy Commissioner, Devanahalli,

Doddaballapura, Bangalore – 562110

Employee Burnout Prediction Using Machine Learning

Project Live Link

GitHub Repository: https://github.com/kano8689/Employee-Burnout

Live Application: https://employeeburnoutml.vercel.app/

1. Abstract

Employee burnout has become one of the most critical challenges faced by organizations across industries. Increasing workloads, tight deadlines, remote working environments, and poor work-life balance contribute significantly to physical and mental exhaustion among employees. Burnout negatively impacts employee productivity, organizational performance, job satisfaction, and retention rates. Traditional methods of identifying burnout, such as employee surveys, manager observations, and periodic performance reviews, often rely on subjective judgment and fail to detect burnout at an early stage. Therefore, organizations require intelligent, data-driven solutions capable of identifying burnout risks accurately and efficiently.

This project presents an Artificial Intelligence and Machine Learning-based Employee Burnout Prediction System that predicts an employee's burnout score using workplace and behavioral attributes. The proposed solution employs the Random Forest Regression algorithm to estimate a continuous burnout score ranging from 0 to 1. Based on the predicted score, employees are automatically categorized into Low, Medium, or High burnout risk levels. Along with prediction, the application also provides recommendations and feature importance analysis, enabling users to understand which workplace factors contribute most to burnout.

The system follows a complete Machine Learning pipeline consisting of data collection, preprocessing, feature engineering, model training, evaluation, deployment, and real-time prediction. Data preprocessing techniques such as missing value handling, categorical feature encoding, feature scaling, and feature selection ensure that the model receives clean and structured input data. The trained Random Forest model demonstrates strong predictive capability while maintaining robustness against noisy data and overfitting. Performance is evaluated using standard regression metrics including R^2 Score, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

To make the model practical and accessible, it is deployed as a web application using the Flask framework. The application supports both single-employee prediction through an interactive form and bulk prediction through CSV file uploads. Users receive real-time predictions accompanied by an animated speedometer gauge, burnout classification, feature contribution percentages, and personalized recommendations. Additionally, a dedicated Model Insights page displays evaluation metrics, feature importance rankings, and dataset information to improve transparency and confidence in the model.

Overall, this project demonstrates how Machine Learning can be effectively integrated with modern web technologies to support Human Resource Analytics. The developed application provides an intelligent decision-support tool that enables organizations to identify employees at risk of burnout proactively, allowing timely intervention and promoting a healthier, more productive workplace. The system also serves as a scalable foundation for future enhancements such as Explainable AI (XAI), cloud integration, and real-time HR database connectivity. The report builds on the existing project documentation while incorporating the latest application features and architecture.

2. Introduction

Employee well-being has emerged as a key determinant of organizational success in today's competitive business environment. While organizations increasingly emphasize productivity and performance, employees often face high workloads, continuous deadlines, extended working hours, and growing workplace pressure. Over time, these conditions can lead to employee burnout—a state of emotional, physical, and mental exhaustion that reduces motivation, job satisfaction, and overall performance. If left unaddressed, burnout may result in decreased productivity, increased absenteeism, higher employee turnover, and significant financial losses for organizations.

Traditionally, organizations rely on employee feedback surveys, managerial observations, counseling sessions, or periodic performance evaluations to assess burnout. Although these methods provide valuable insights, they are time-consuming, subjective, and often incapable of detecting burnout before it significantly affects employee performance. Human judgment may vary across managers, and manual evaluation becomes increasingly difficult as organizations grow larger. Consequently, many employees remain unnoticed until burnout has already impacted their productivity and well-being.

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have transformed the way organizations analyze employee-related data. Instead of relying solely on human observations, ML algorithms can identify hidden relationships between multiple workplace factors and employee burnout by learning from historical data. These predictive models provide objective, data-driven insights that enable organizations to identify at-risk employees before severe burnout occurs. Early detection allows Human Resource (HR) departments to implement appropriate interventions such as workload balancing, flexible work arrangements, wellness programs, or counseling support.

The Employee Burnout Prediction Using Machine Learning project aims to address these challenges by developing a smart web-based prediction system capable of estimating employee burnout accurately. The system utilizes workplace attributes such as gender, company type, work-from-home availability, designation level, resource allocation, and mental fatigue score to predict a continuous burnout score between 0 and 1. Unlike traditional classification systems that only identify risk categories, this approach provides both an exact burnout score and meaningful burnout classifications (Low, Medium, or High), offering greater flexibility and interpretability.

The core prediction engine is built using the Random Forest Regression algorithm, an ensemble Machine Learning technique known for its high prediction accuracy, robustness against noisy data, and ability to estimate feature importance. One of the major advantages of Random Forest is its capability to determine how strongly each input feature contributes to the final prediction. This functionality enables the application to present users with a Feature Impact (%) visualization, making the prediction process more transparent and understandable.

To ensure practical usability, the Machine Learning model is integrated into a Flask-based web application featuring an intuitive and responsive user interface. The application supports both individual employee prediction through an interactive web form and bulk prediction through CSV file uploads, allowing HR professionals to evaluate large employee datasets efficiently. Additional features such as an animated burnout speedometer, personalized recommendations, model evaluation metrics, and a Model Insights dashboard further enhance the user experience.

The project follows a complete end-to-end Machine Learning workflow, beginning with data preprocessing and feature engineering, followed by model training, evaluation, deployment, and real-time prediction. It demonstrates how modern AI technologies can be applied to solve real-world Human Resource Analytics problems while maintaining scalability, usability, and prediction transparency. Compared to conventional burnout assessment methods, the proposed system significantly reduces manual effort, improves prediction consistency, and provides actionable insights for organizational decision-making.

Furthermore, the project has been successfully deployed online using Flask and Vercel, making it accessible through any modern web browser without requiring local installation. This deployment highlights the practical applicability of Machine Learning models beyond experimental environments and demonstrates how AI-powered decision-support systems can assist organizations in maintaining healthier, more productive workforces.

3. Methodology

The Employee Burnout Prediction System follows a structured Machine Learning methodology that transforms raw employee information into meaningful burnout predictions. The methodology consists of multiple stages, including data collection, preprocessing, feature engineering, model training, evaluation, prediction, and deployment. Each stage is designed to improve prediction accuracy while ensuring the system remains reliable, scalable, and easy to use. By following this systematic approach, the developed application provides accurate burnout predictions that can support Human Resource (HR) departments in making informed decisions regarding employee well-being.

The first stage of the methodology involves **data collection**. The model is trained using an employee burnout dataset containing demographic and workplace-related attributes that influence employee stress levels. Each record represents an individual employee and contains information such as Gender, Company Type, Work From Home (WFH) availability, Designation, Resource Allocation, Mental Fatigue Score, and Burn Rate. The Burn Rate is the target variable that represents the burnout level of an employee as a continuous numerical value between 0 and 1. These features were selected because they have a direct or indirect relationship with employee workload, workplace conditions, and psychological stress.

After collecting the dataset, **data preprocessing** is performed to prepare the data for Machine Learning. Real-world datasets often contain missing values, inconsistent formats, duplicate records, or categorical values that cannot be processed directly by Machine Learning algorithms. Therefore, preprocessing plays an essential role in improving model performance. The application validates the dataset before training and removes or handles missing values wherever necessary. Categorical variables such as Gender, Company Type, and WFH Setup Availability are converted into numerical representations using **Label Encoding**. This transformation enables the Random Forest algorithm to interpret categorical information while preserving relationships between different categories.

The next preprocessing step is **feature scaling**. Since numerical attributes such as Designation, Resource Allocation, and Mental Fatigue Score have different numerical ranges, they are standardized using **StandardScaler**. Standardization transforms numerical values into a common scale with zero mean and unit variance, preventing features with larger numerical values from dominating the learning process. Although Random Forest algorithms are generally less sensitive to feature scaling than many other Machine Learning models, standardization ensures consistency throughout the preprocessing pipeline and improves compatibility with future model enhancements.

Once preprocessing is completed, the dataset is divided into **training and testing subsets** using an 80:20 split. Approximately 80% of the dataset is used to train the Machine Learning model, while the remaining 20% is reserved for evaluating its performance on previously unseen data. This approach helps determine whether the trained model has generalized effectively rather than memorizing the training data. Proper train-test splitting reduces the risk of overfitting and provides a realistic estimate of prediction performance in practical applications.

Following preprocessing, the project enters the **feature engineering** stage. Rather than including every available attribute, only the most relevant workplace factors are selected for prediction. The final feature set includes Gender, Company Type, WFH Setup Availability, Designation Level, Resource Allocation, and Mental Fatigue Score. These attributes collectively provide sufficient information for estimating employee burnout while maintaining a relatively simple and interpretable prediction model. Selecting meaningful features also reduces computational complexity and improves prediction efficiency.

The core prediction engine of the application is based on the **Random Forest Regression algorithm**, a supervised ensemble Machine Learning technique. Unlike a single Decision Tree, Random Forest constructs numerous decision trees using randomly selected subsets of training samples and features. During prediction, the outputs of all trees are combined to produce a final burnout score. This ensemble approach significantly improves prediction accuracy while reducing variance and minimizing the risk of overfitting. Random Forest is particularly suitable for this application because it performs well on mixed numerical and categorical datasets, handles noisy data effectively, and provides built-in feature importance analysis.

One of the major advantages of the selected algorithm is its ability to calculate **Feature Importance**. Instead of simply predicting a burnout score, the model determines how much each input feature contributes to the final prediction. The calculated importance values are converted into percentage contributions and displayed within the web application as **Factor Impact (%)**. This functionality increases transparency by helping users understand why the model produced a particular prediction. In most cases, the Mental Fatigue Score contributes the highest percentage toward burnout prediction, followed by Resource Allocation and Designation Level, indicating that workload and mental stress have the strongest influence on employee burnout.

After model training, the Random Forest model is evaluated using three standard regression metrics: **Coefficient of Determination (R^2 Score), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE)**. The R^2 Score measures how effectively the model explains variations in burnout scores, while RMSE and MAE quantify prediction errors. Lower RMSE and MAE values indicate better predictive performance. These evaluation metrics provide confidence that the trained model performs accurately on unseen employee records and can be reliably used for real-world prediction tasks.

Once the evaluation process is complete, the trained Machine Learning model, along with the Label Encoders, StandardScaler parameters, and evaluation metrics, is stored using serialization techniques. Saving these components eliminates the need to retrain the model every time the application starts, thereby reducing startup time and improving system efficiency. During application initialization, the saved model and preprocessing objects are loaded automatically into memory, making real-time predictions possible within milliseconds.

The final stage of the methodology involves **deployment and real-time prediction**. The Machine Learning model is integrated with a Flask-based web application that serves as the user interface. Users can either manually enter employee information for single predictions or upload CSV files containing multiple employee records for batch processing. Submitted data is automatically validated, encoded, scaled, and passed to the trained Random Forest model. The predicted burnout score is then categorized into Low (0.00–0.30), Medium (0.31–0.60), or High (0.61–1.00) burnout levels. The application displays the burnout score using an animated speedometer gauge, presents feature contribution percentages, and generates personalized recommendations based on the predicted risk level.

This end-to-end methodology demonstrates the integration of data preprocessing, feature engineering, Machine Learning, model evaluation, explainability, and web deployment into a single intelligent application. The structured workflow ensures prediction accuracy, user transparency, and scalability, making the developed system suitable for educational demonstrations as well as practical Human Resource Analytics applications. The methodology also provides a strong foundation for future improvements, including Explainable Artificial Intelligence (XAI), cloud deployment, continuous model retraining, and integration with enterprise Human Resource Management Systems (HRMS).

4. Implementation & Results

The Employee Burnout Prediction System has been implemented as a complete end-to-end Machine Learning web application that combines data preprocessing, model training, prediction, visualization, and deployment into a single platform. The application is developed using **Python** as the primary programming language, while the **Flask** framework is used to build the backend server and manage communication between the Machine Learning model and the user interface. The frontend is designed using **HTML5**, **CSS3**, and **JavaScript**, providing a responsive and interactive environment for users. The project has also been deployed online through **Vercel**, allowing users to access the application directly from a web browser without requiring local installation.

During application startup, the system first checks whether a trained Machine Learning model already exists. If a previously trained model is available, it is loaded directly into memory together with the saved Label Encoders, StandardScaler parameters, and evaluation metrics. If these files are unavailable, the application automatically loads the dataset, performs preprocessing, trains a new Random Forest Regression model, evaluates its performance, and saves the generated model for future use. This automatic model loading mechanism significantly reduces execution time and improves the efficiency of subsequent predictions.

The application provides two prediction modes to satisfy different organizational requirements. The **Single Employee Prediction** module allows users to manually enter employee details, including Gender, Company Type, Work From Home availability, Designation Level, Resource Allocation, and Mental Fatigue Score. After validating the input data, the application performs feature encoding and scaling before passing the processed information to the trained Random Forest model. The model generates a burnout score between 0 and 1, which is then classified into Low, Medium, or High burnout categories. Along with the predicted burnout score, the application displays an animated speedometer gauge, personalized recommendations, and a detailed Feature Impact (%) chart showing how each feature contributed to the prediction. This explainable prediction process improves user confidence by making the Machine Learning model more transparent.

The second major functionality is the **Bulk CSV Prediction** module, which is designed for Human Resource departments that need to evaluate multiple employees simultaneously. Instead of manually entering employee details one at a time, users can upload a CSV file containing hundreds of employee records. The application automatically validates the uploaded dataset, preprocesses every record, performs predictions for all employees, and generates a comprehensive summary dashboard. The results include the total number of employees analyzed, average burnout score, distribution of burnout categories, identification of high-risk employees, and a detailed prediction table. A Top-5 High-Risk Employees section with miniature speedometer gauges further assists HR professionals in prioritizing employees who require immediate attention. This module significantly reduces manual effort and enables efficient workforce analysis.

Another important component of the application is the **Model Insights Dashboard**. Unlike many prediction systems that only display final results, this dashboard provides technical information regarding the trained Machine Learning model. Users can view important regression evaluation metrics such as the **Coefficient of Determination (R^2 Score), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE)**. In addition, the dashboard presents feature importance rankings, dataset information, training and testing sample sizes, and the list of input features used during model training. These insights improve transparency, allowing users to better understand how the prediction model operates and evaluate its reliability.

The developed application successfully demonstrates the integration of Machine Learning with modern web technologies. The prediction process is completed within a few seconds, providing immediate feedback to users while maintaining prediction consistency. The responsive user interface ensures compatibility across desktop and mobile devices, making the system suitable for practical organizational use as well as academic demonstrations. Overall, the implementation showcases how Artificial Intelligence can be effectively deployed through a user-friendly web application to support data-driven decision-making in Human Resource Analytics.

The overall performance of the Random Forest Regression model indicates that the selected algorithm is well suited for predicting employee burnout. By combining multiple decision trees, the model produces stable predictions while minimizing overfitting. Furthermore, the built-in feature importance capability enables explainability without requiring additional computational techniques. The successful deployment of the application demonstrates that Machine Learning models can be integrated seamlessly into real-world software systems, providing organizations with practical decision-support tools capable of improving employee well-being and organizational productivity.

5. Conclusion

Employee burnout is a growing concern for organizations worldwide because it directly influences employee productivity, job satisfaction, workplace engagement, and overall organizational performance. Traditional approaches for identifying burnout often depend on manual evaluations, employee surveys, and managerial observations, which are subjective, time-consuming, and difficult to scale. The Employee Burnout Prediction System developed in this project demonstrates how Artificial Intelligence and Machine Learning can provide an intelligent, objective, and efficient alternative for identifying burnout risks at an early stage.

The developed application successfully integrates data preprocessing, feature engineering, Random Forest Regression, model evaluation, and web deployment into a unified end-to-end Machine Learning solution. By analyzing workplace-related factors such as Gender, Company Type, Work From Home availability, Designation, Resource Allocation, and Mental Fatigue Score, the system predicts a continuous burnout score ranging from 0 to 1 and classifies employees into Low, Medium, or High burnout categories. This approach provides more detailed information than simple classification models, enabling HR professionals to better understand the severity of employee burnout.

One of the key strengths of this project is its emphasis on **model explainability**. Instead of producing only prediction results, the application also presents Feature Impact (%) analysis, allowing users to understand which employee attributes contribute most significantly to burnout. The inclusion of an animated speedometer gauge, personalized recommendations, bulk CSV prediction, and a dedicated Model Insights dashboard further enhances the usability and transparency of the application. These features make the system not only technically effective but also user-friendly for HR professionals who may not possess extensive Machine Learning knowledge.

The successful deployment of the application using Flask and Vercel demonstrates that Machine Learning models can be transformed into practical web-based decision-support systems. Real-time prediction capability, responsive user interface design, and support for batch employee analysis make the application suitable for educational purposes as well as real-world organizational environments. The project also highlights the importance of combining data science techniques with software engineering principles to develop scalable AI-powered applications.

Although the current system provides reliable burnout predictions, there remains considerable scope for future enhancement. Integration with real-time Human Resource Management Systems (HRMS), cloud databases, Explainable Artificial Intelligence (XAI) techniques such as SHAP or LIME, user authentication, continuous model retraining, and advanced analytics dashboards could further improve the accuracy, scalability, and interpretability of the application. Overall, this project demonstrates the practical application of Machine Learning in Human Resource Analytics and provides a strong foundation for building intelligent employee wellness monitoring systems capable of supporting healthier, more productive workplaces.

6. References

1. Breiman, L. (2001). *Random Forests*. Machine Learning, 45(1), 5–32.
2. Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.
3. Python Software Foundation. *Python Documentation*. Available at: https://docs.python.org/
4. Flask Documentation. Available at: https://flask.palletsprojects.com/
5. Scikit-learn Documentation. Available at: https://scikit-learn.org/
6. NumPy Documentation. Available at: https://numpy.org/
7. Pandas Documentation. Available at: https://pandas.pydata.org/
8. Vercel Documentation. Available at: https://vercel.com/docs
9. Krishnam Mavani. *Employee Burnout Prediction Using Machine Learning* (GitHub Repository). https://github.com/kano8689/Employee-Burnout
10. Live Project: https://employeeburnoutml.vercel.app/